

CHAPTER 04

Waves

From the song stuck in your head, to the Wi-Fi connecting your phone, to the colors you see in a sunset — waves are quietly running your entire life. Time to learn what they really are and how they work.



Georgia Standard S8P4

WHAT YOU'LL LEARN

- | | |
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| 01 What Are Waves? | 06 Wave Speed & Media |
| 02 Mechanical vs. EM Waves | 07 The EM Spectrum |
| 03 Transverse vs. Longitudinal | 08 Sound Waves |
| 04 Wave Properties | 09 Wave Behaviors |
| 05 Properties & Energy | 10 Lenses & Light |

4.1 What Are Waves?

Throw a rock into a still pond. Watch what happens — ripples spread outward in circles, traveling away from where the rock landed. But here's the wild part: **the water itself isn't going anywhere**. A leaf floating on the pond just bobs up and down as the ripples pass underneath. The pattern moves. The water doesn't.

That's the secret to waves. A **wave** is a disturbance that **transfers energy from one place to another without transferring matter**. The wave moves. The stuff the wave is moving through stays roughly where it started. Sound waves move energy through air, but the air itself doesn't travel from the speaker to your ear. Light waves carry energy across the universe, but they're not carrying any "stuff." Waves are how the universe moves energy around without actually shipping particles all over the place.

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Did You Know?

When a stadium crowd does "the wave," nobody actually moves around the stadium. Each fan just stands up and sits down. But the **pattern** travels all the way around the bowl. That's exactly how every wave in physics works — patterns move, particles stay put.

★

Real-World Connection

The energy from the Sun travels 93 million miles of empty space to reach Earth — as a wave. Without that wave delivering energy from the Sun, no plants would grow, no weather would happen, and Earth would be a frozen rock. Waves keep our planet alive.

4.2 Mechanical vs. Electromagnetic Waves

All waves fall into two huge categories — based on whether they need "stuff" to travel through.

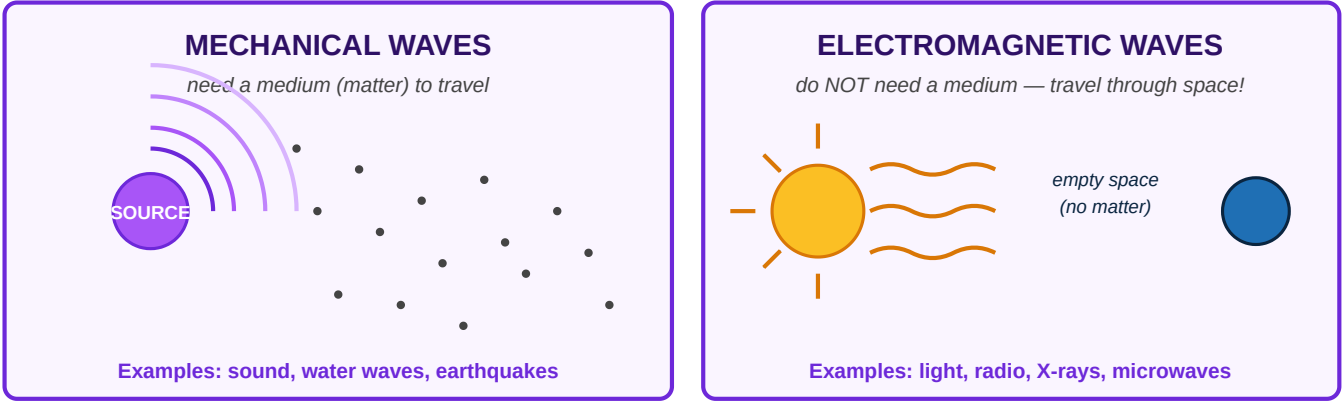


Figure 4.1 — Mechanical waves need a medium (like air or water). Electromagnetic waves can travel through empty space.

Similarities & Differences

Feature	Mechanical Waves	Electromagnetic Waves
Need a medium?	YES (air, water, solid, etc.)	NO — can travel in vacuum
Examples	Sound, water waves, seismic	Light, radio, microwaves, X-rays
Speed in air	Slow (sound $\approx$ 343 m/s)	Very fast (light $\approx$ 300,000,000 m/s)
Carry energy?	YES	YES
Transfer matter?	NO	NO

Common Mix-Up

If you were in space (no air) and an asteroid exploded right next to you, you would **see** the flash but **not hear** the boom. Light is electromagnetic and travels through the vacuum just fine. Sound is mechanical and needs molecules to vibrate — no air, no sound. Despite what movies show, space is silent.

4.3 Transverse vs. Longitudinal Waves

Waves can also be sorted by **which direction the particles vibrate** compared to the direction the wave is traveling.

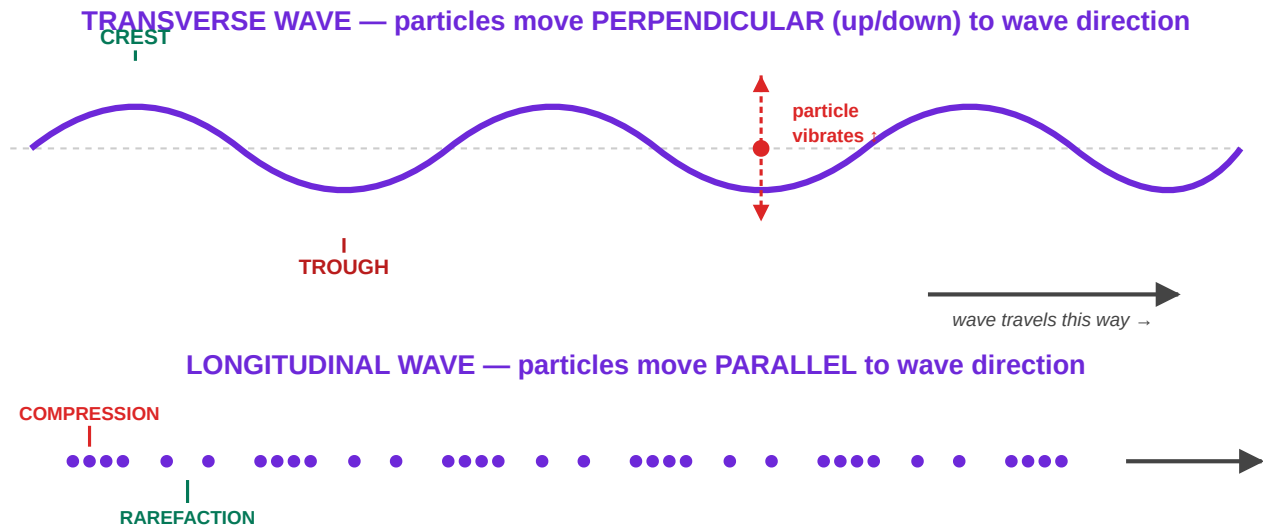


Figure 4.2 — In a transverse wave, particles move side-to-side. In a longitudinal wave, particles squish together and spread apart along the wave's path.

Parts of a Transverse Wave

- **Crest:** The **highest** point of the wave.
- **Trough:** The **lowest** point of the wave.

Parts of a Longitudinal Wave

- **Compression:** The part where particles are **squeezed close together**.
- **Rarefaction:** The part where particles are **spread far apart**.

Try It Yourself

Grab a Slinky. Shake one end side-to-side — you'll see a **transverse wave** ripple down the spring. Now push and pull one end straight along the Slinky's length — you'll see **compressions and rarefactions** moving down the spring. That's a longitudinal wave. Sound works exactly like that second one.

4.4 Wave Properties

Every wave has measurable properties. Knowing the vocabulary lets you describe and compare any wave on Earth.

Anatomy of a Wave

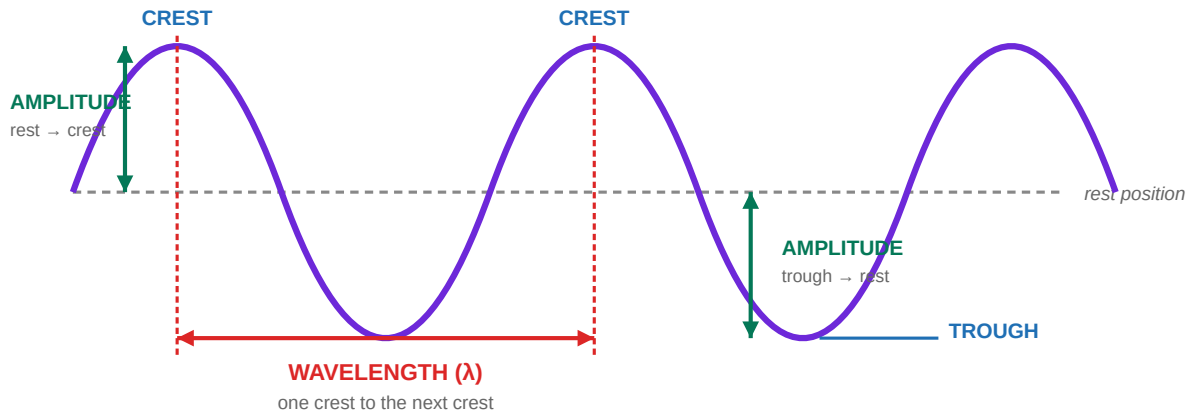


Figure 4.3 — Wavelength is measured from one crest to the next crest. Amplitude is the distance from rest position to a crest (or from a trough back to rest position).

The Three Big Properties

Property	Definition	Units
Wavelength (λ)	Distance from one crest to the next (or compression to next compression)	meters (m)
Frequency (f)	Number of waves that pass a point per second	Hertz (Hz) — 1 Hz = 1 wave per second
Amplitude	Height of the wave from its rest position	meters (m) or decibels for sound

! Did You Know?

The radio station 94.9 FM broadcasts at a frequency of 94.9 **MHz** — that's 94,900,000 waves passing your antenna every single second. Your phone's Wi-Fi uses 2.4 GHz — that's 2,400,000,000 waves per second. Mind-blowing speeds, but your devices handle them effortlessly.

4.5 Wave Properties & Energy

Wave properties aren't just for math class — they directly determine how much **energy** a wave carries.

High vs. Low Energy Waves

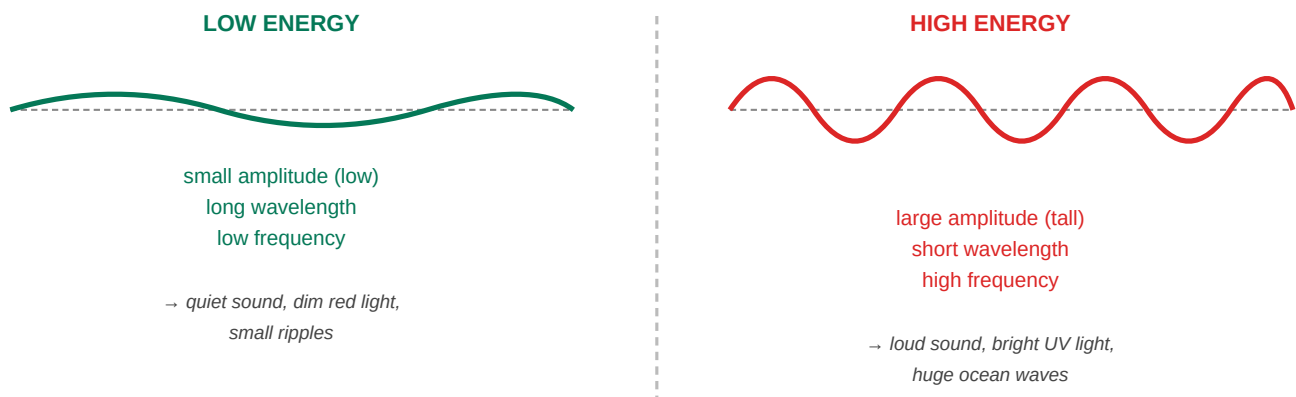


Figure 4.4 — Bigger amplitude = more energy. Higher frequency (and shorter wavelength) = more energy.

→ The Energy Rules

- **Higher amplitude = more energy.** A loud song has more energy than a quiet one. A tsunami has way more energy than a tiny ripple.
- **Higher frequency = more energy.** Gamma rays (super high frequency) are way more energetic than radio waves (low frequency). UV light has more energy than visible light.
- **Shorter wavelength = more energy.** Frequency and wavelength are linked — when one goes up, the other goes down.

The Frequency-Wavelength Relationship

Frequency and wavelength have an **inverse relationship**: when one increases, the other decreases. Think of it like fitting waves into a fixed space. If your waves are long, only a few fit. If you make them shorter, more can squeeze in — meaning more pass by per second (higher frequency).

4.6 Wave Speed & Density of Media

How fast a wave travels depends on the **medium** it's traveling through — and surprisingly, the rules for sound and light are *opposite* of each other.

Sound (Mechanical Waves): Denser = Faster

Sound needs particles to bump into each other and pass along the vibration. The closer the particles are packed, the faster they can pass the wave along. Sound travels:

Medium	Density	Speed of Sound
Air	Low density (gas)	~343 m/s
Water	Medium density (liquid)	~1,500 m/s (4× faster!)
Steel	High density (solid)	~5,000 m/s (15× faster!)

This is why if you press your ear to a railroad track, you can hear a train approaching long before you can hear it through the air — the steel rails carry the sound way faster than the air does.

Light (EM Waves): Less Dense = Faster

Light is the opposite. Light travels **fastest in a vacuum** (empty space, no medium at all) at 300,000,000 m/s. It slows down when it has to pass through matter — and the denser the matter, the slower it goes. Light moves slightly slower through air, slower through water, and even slower through glass or diamond.

✗ Common Mix-Up

Sound and light follow **opposite rules** when it comes to density. Sound speeds up in denser materials. Light slows down in denser materials. That's why a thunderstorm flash arrives *before* the thunder — light travels through air almost instantly, but sound takes a few seconds to reach you.

? Quick Check 4.6

1. You see a lightning flash, then count "one Mississippi" before hearing the thunder. Why is there a delay?
2. Why can dolphins communicate over miles in the ocean using sound?
3. Predict: would sound travel faster through Jello or through ice? Why?

4.7 The Electromagnetic Spectrum

Visible light is just a tiny sliver of a much bigger family of waves called the **electromagnetic spectrum**. All EM waves are the same kind of thing — they just have different wavelengths, frequencies, and energies.

The Electromagnetic Spectrum

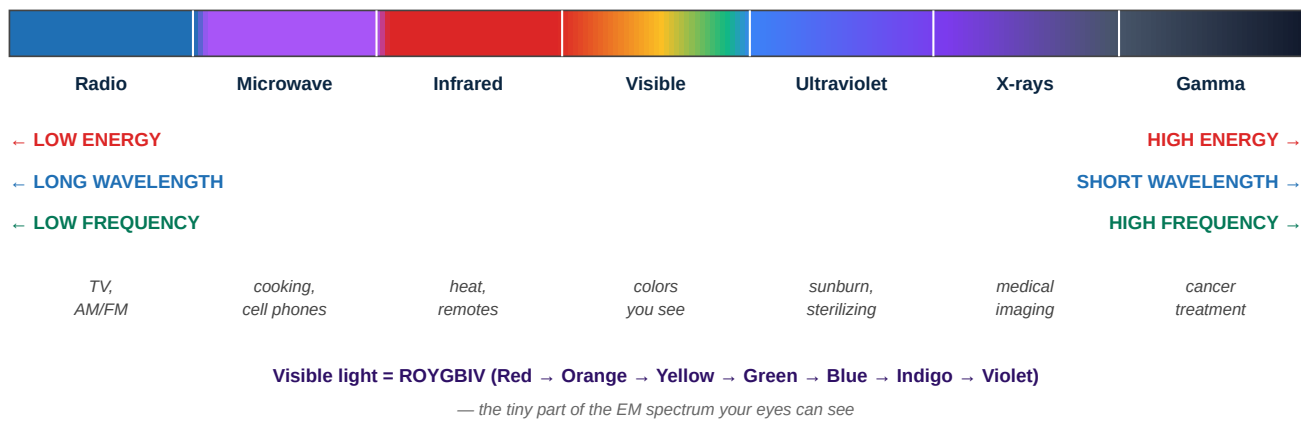


Figure 4.5 — The full EM spectrum. The only part your eyes can see is the tiny visible light slice in the middle.

From Lowest to Highest Energy

1. **Radio Waves** — Used for AM/FM radio, broadcast TV, walkie-talkies.
2. **Microwaves** — Heat your leftovers, carry your cell signal, beam data to satellites.
3. **Infrared** — Heat radiation; what TV remotes use; night-vision goggles see it.
4. **Visible Light** — The colors of the rainbow. The **ONLY** waves your eyes detect.
5. **Ultraviolet (UV)** — Gives you sunburns; sterilizes medical equipment; what bug zappers use.
6. **X-rays** — Pass through skin and muscle but not bone — perfect for medical imaging.
7. **Gamma Rays** — The most energetic of all; used to treat cancer; produced by nuclear reactions.

★ Real-World Connection

Pull out your phone — it uses the EM spectrum in tons of ways. Wi-Fi and cell signal? Microwaves. The screen lighting up? Visible light. The IR sensor that locks your phone when you're not looking? Infrared. The Face ID dots that map your face? Infrared again. Even your phone's GPS works because satellites send **microwave signals** down from orbit.

! Did You Know?

The microwave oven was invented **by accident** in 1945. A scientist named Percy Spencer was working with a radar device when he noticed a chocolate bar in his pocket had melted. He realized the microwaves were heating it up — and soon after, microwave ovens were a thing. Today they're in 90% of American kitchens.

4.8 Sound Waves

Sound is a **longitudinal mechanical wave**. It's produced when something vibrates — a guitar string, vocal cords, a speaker cone — and pushes air molecules around in compressions and rarefactions. Those vibrations spread through the air, and when they hit your eardrum, they make it vibrate too. Your brain translates those vibrations into what you "hear."

Two Important Sound Concepts

→ Pitch = Frequency

Pitch is how **high** or **low** a sound is. It depends on the wave's frequency.

- High frequency → high pitch (whistle, squeaky voice)
- Low frequency → low pitch (bass drum, deep voice)

→ Volume = Amplitude

Volume (or loudness) is how **loud** or **soft** a sound is. It depends on the wave's amplitude.

- Large amplitude → loud sound
- Small amplitude → quiet sound

! Did You Know?

Humans hear sound frequencies between about **20 Hz and 20,000 Hz**. Dogs can hear up to 45,000 Hz (which is why dog whistles work — we can't hear them, but dogs can). Bats hunt using sounds up to 200,000 Hz, and elephants communicate in the opposite direction — using rumbles below 20 Hz that travel for miles through the ground.

4.9 Wave Behaviors

What happens when a wave hits a new material? It can do five different things — and most of the time, it does several at once.

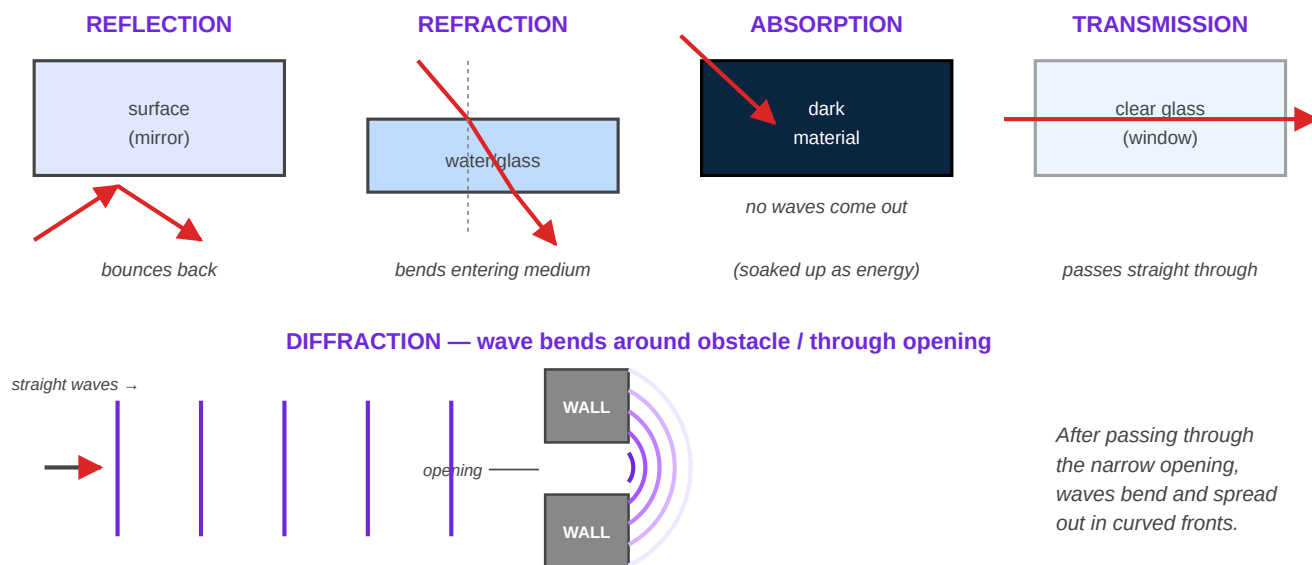


Figure 4.6 — The five behaviors waves can do when they meet a new material or obstacle.

Behavior	What Happens	Real Example
Reflection	Wave bounces off a surface	Looking in a mirror; an echo in a canyon
Refraction	Wave bends when entering a new medium	A straw "broken" in a glass of water; rainbows
Absorption	Wave's energy is taken in by a material	Black clothing getting hot in sunlight
Diffraction	Wave bends around an obstacle or through a gap	Hearing someone talking in the next room
Transmission	Wave passes through a material	Light passing through a clear window

Three Types of Materials by How They Treat Light

- **Transparent:** Light passes through clearly. You can see through it. (window glass, clean water)
- **Translucent:** Light passes through, but gets scattered. You can see light but not details. (frosted glass, wax paper)
- **Opaque:** Light cannot pass through at all. (a brick wall, your textbook)

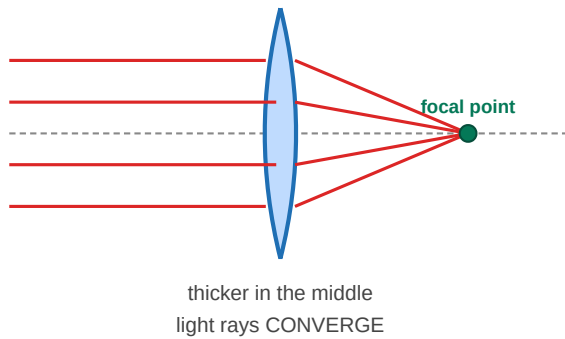
? Quick Check 4.9

1. Identify the behavior: (a) a flashlight beam shines through a glass of water, (b) you hear your friend talking around a corner, (c) a pool's bottom looks shallower than it actually is.
2. Why does a black T-shirt feel hotter in the sun than a white T-shirt?

4.10 Lenses & Light

A **lens** is a piece of clear, curved material — usually glass or plastic — that **refracts (bends) light** to form an image. There are two main types of lenses, and they bend light in opposite ways.

CONVEX LENS (Converging)



CONCAVE LENS (Diverging)

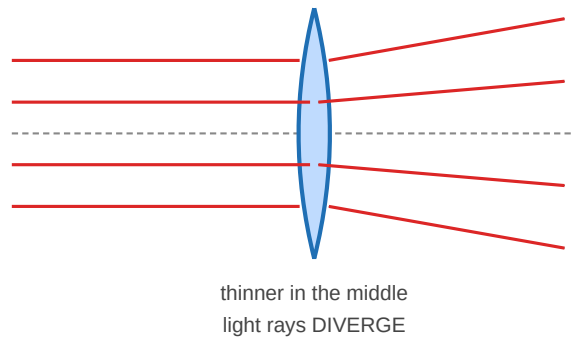


Figure 4.7 — Convex lenses pull light together. Concave lenses spread light apart.

Convex Lenses (Converging)

A **convex lens** is **thicker in the middle** than at the edges. It bulges out. Light rays passing through it bend **toward a single point** called the focal point. Convex lenses form magnified or focused images.

Uses: Magnifying glasses, eyeglasses for farsighted people, camera lenses, the lens in your eyeball, telescopes.

Concave Lenses (Diverging)

A **concave lens** is **thinner in the middle** than at the edges. It caves inward. Light rays passing through it **spread apart**. Concave lenses make objects appear smaller and farther away.

Uses: Eyeglasses for nearsighted people, peepholes in doors, some flashlights.

★ Real-World Connection

Your eye contains a built-in **convex lens**. It bends incoming light to focus a tiny upside-down image onto the back of your eye (the retina). If your lens is shaped wrong, you'll need glasses — convex glasses for farsightedness, concave glasses for nearsightedness. Either way, you're stacking another lens on top of the one nature gave you.

! Did You Know?

The Hubble Space Telescope and the newer James Webb Space Telescope both use giant lens-and-mirror systems to focus light from galaxies trillions of miles away. The James Webb's main mirror is over 21 feet wide — about the size of a school bus — and it's letting astronomers see stars that formed less than 200 million years after the Big Bang.

? Quick Check 4.10

1. Which type of lens would you use as a magnifying glass? Why?
2. What does it mean for a lens to "converge" light? What does "diverge" mean?
3. If someone is nearsighted, what shape of lens corrects their vision?

★ CHAPTER 4 BIG IDEAS

- A **wave** transfers energy without transferring matter.
- **Mechanical waves** (sound) need a medium. **Electromagnetic waves** (light) can travel through empty space.
- **Transverse waves**: particles move perpendicular to wave direction (light, water). **Longitudinal waves**: particles move parallel (sound).
- Key wave properties: **wavelength** (m), **frequency** (Hz), **amplitude** (height).
- **Higher amplitude OR higher frequency = more energy**. Frequency and wavelength are inversely related.
- Sound travels FASTER in denser media. Light travels SLOWER in denser media. (Opposite rules!)
- The **EM spectrum** from low to high energy: radio → microwave → infrared → visible → UV → X-rays → gamma rays.
- Sound waves: **pitch** = frequency, **volume** = amplitude.
- Wave behaviors: **reflection, refraction, absorption, diffraction, transmission**.
- **Convex lenses** converge light (magnifiers, eyes). **Concave lenses** diverge light (peepholes).

I Can... (Self-Check)

- ☐ I can explain similarities and differences between electromagnetic and mechanical waves. (S8P4.a)
- ☐ I can use data to illustrate the relationship between the electromagnetic spectrum and energy. (S8P4.b)
- ☐ I can describe practical applications of the EM spectrum (communication, medical, military). (S8P4.c)
- ☐ I can compare how light and sound waves are reflected, refracted, absorbed, diffracted, or transmitted. (S8P4.d)
- ☐ I can predict patterns in the relationship between density of media and wave speed. (S8P4.e)
- ☐ I can describe relationships between wave properties (frequency, amplitude, wavelength) and energy. (S8P4.f)
- ☐ I can demonstrate the effects lenses have on light and describe their technological applications. (S8P4.g)